

Solution to the Exercise proposed in the “Back-propagation Notes”

Alessio Micheli

Direct derivation (alternative) for Case 2: hidden unit j

$$\begin{aligned}\boxed{-\frac{\partial E_p}{\partial o_j}} &= -\frac{\partial}{\partial o_j} \left\{ \frac{1}{2} \sum_{k=1}^K (d_k - o_k)^2 \right\} \\ &= -\sum_{k=1}^K (d_k - o_k) \cdot \frac{\partial(-o_k)}{\partial o_j} \\ &= \sum_{k=1}^K (d_k - o_k) \cdot \frac{\partial f_k(\text{net}_k)}{\partial o_j} \\ &= \sum_{k=1}^K \boxed{(d_k - o_k) f'_k(\text{net}_k)} \cdot \frac{\partial \text{net}_k}{\partial o_j} \\ &= \sum_{k=1}^K \boxed{\delta_k} w_{kj}\end{aligned}$$

where $\boxed{\delta_k}$ is that of Case 1 in the “Back-propagation Notes”.